SOC Alert Triage and Investigation Dashboard Report

Introduction

This report documents the creation and analysis of a Splunk SIEM dashboard used for SOC alert triage and threat investigation. The investigation focused on identifying brute-force authentication attempts, suspicious web activity, persistence indicators, and automated reconnaissance behavior through log correlation and dashboard visualizations.

1. Initial SSH Log Investigation

To begin the investigation process, SSH-related events were searched in Splunk to identify suspicious login activity and authentication attempts. This step helped narrow the focus toward potential brute-force behavior and abnormal access patterns within the environment.

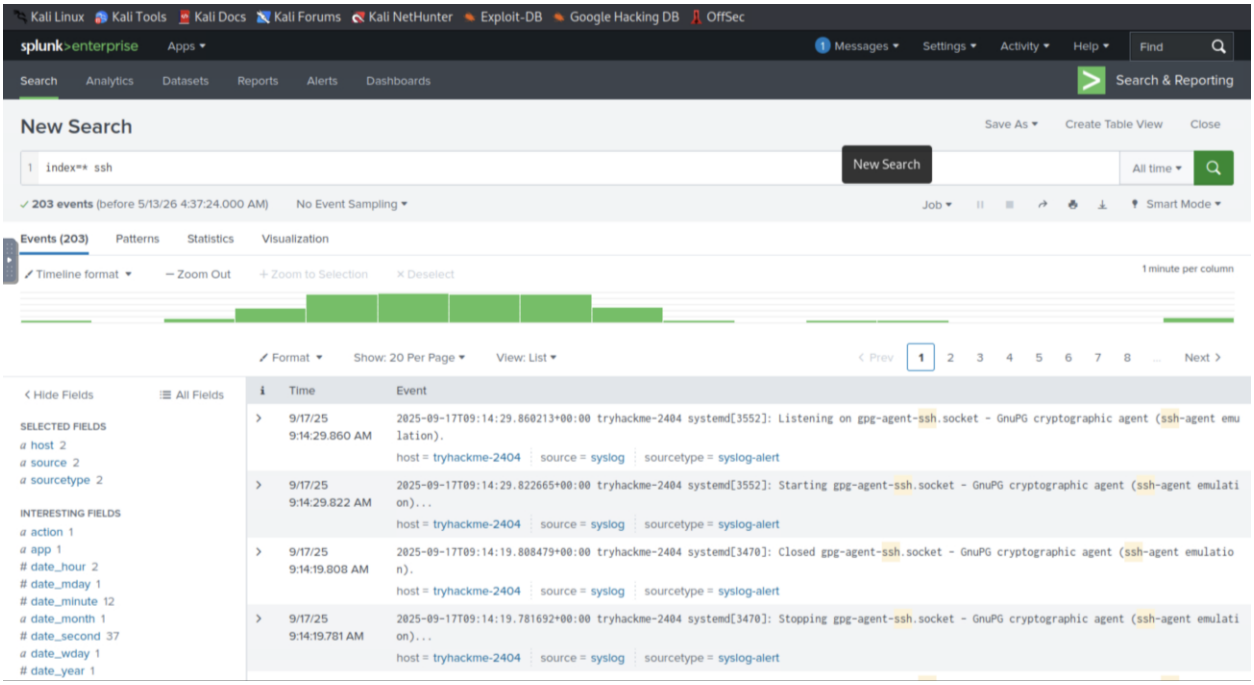

Figure 1. Initial SSH-related event search in Splunk showing authentication and SSH service activity used for early-stage log investigation.

Key Observation:

The SSH keyword search returned over 200 related events, indicating a significant amount of authentication activity that required further filtering and correlation analysis.

2. Identifying Top Attacker IP Addresses

After identifying SSH-related activity, a dashboard visualization was created to determine which source IP addresses generated the highest number of suspicious login attempts. This helped prioritize investigation efforts toward the most active attacker systems observed in the environment.

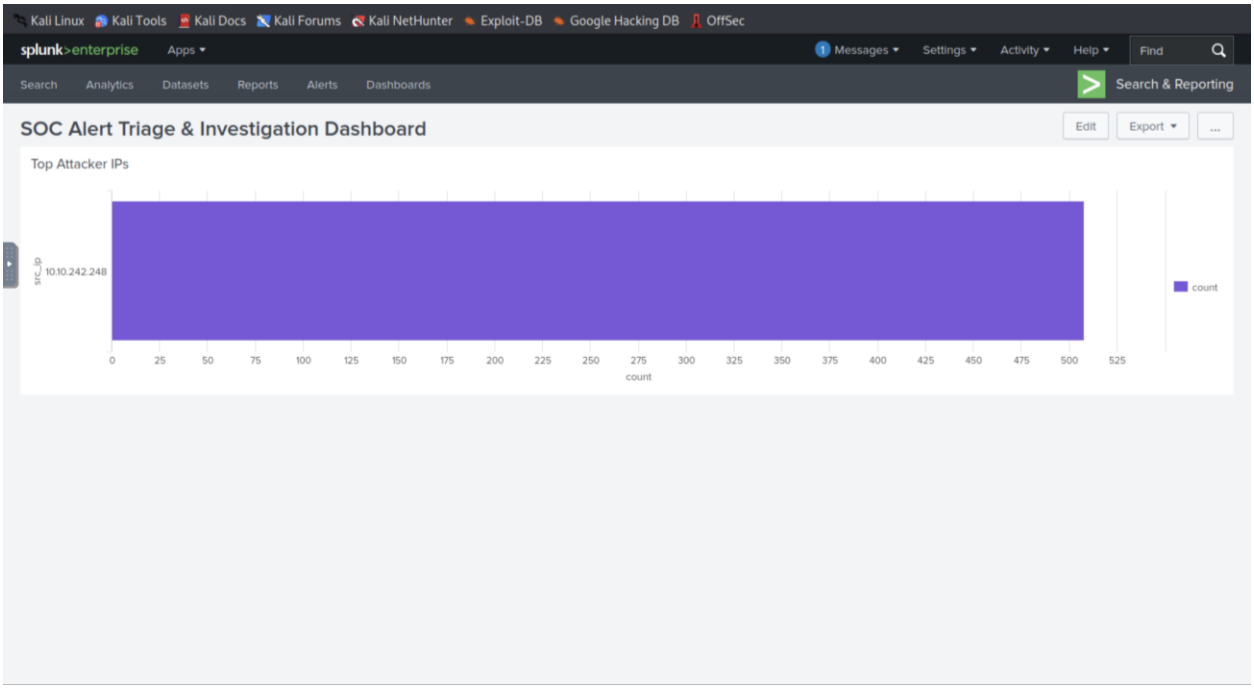

Figure 2. Splunk dashboard visualization displaying the top attacker IP address associated with excessive failed SSH login attempts.

Key Observation:

The source IP address 10.10.242.248 generated more than 500 authentication-related events, indicating possible brute-force or automated attack behavior targeting the SSH service. This visualization improves SOC analyst efficiency by quickly highlighting high-risk attacker sources that may require blocking, escalation, or deeper forensic analysis.

3. Monitoring Failed Login Attempts Over Time

A time-based visualization was added to the dashboard to analyze the frequency and pattern of failed SSH login attempts throughout the investigation period. Tracking login attempts over time helps identify attack spikes, automated behavior, and recurring authentication anomalies.

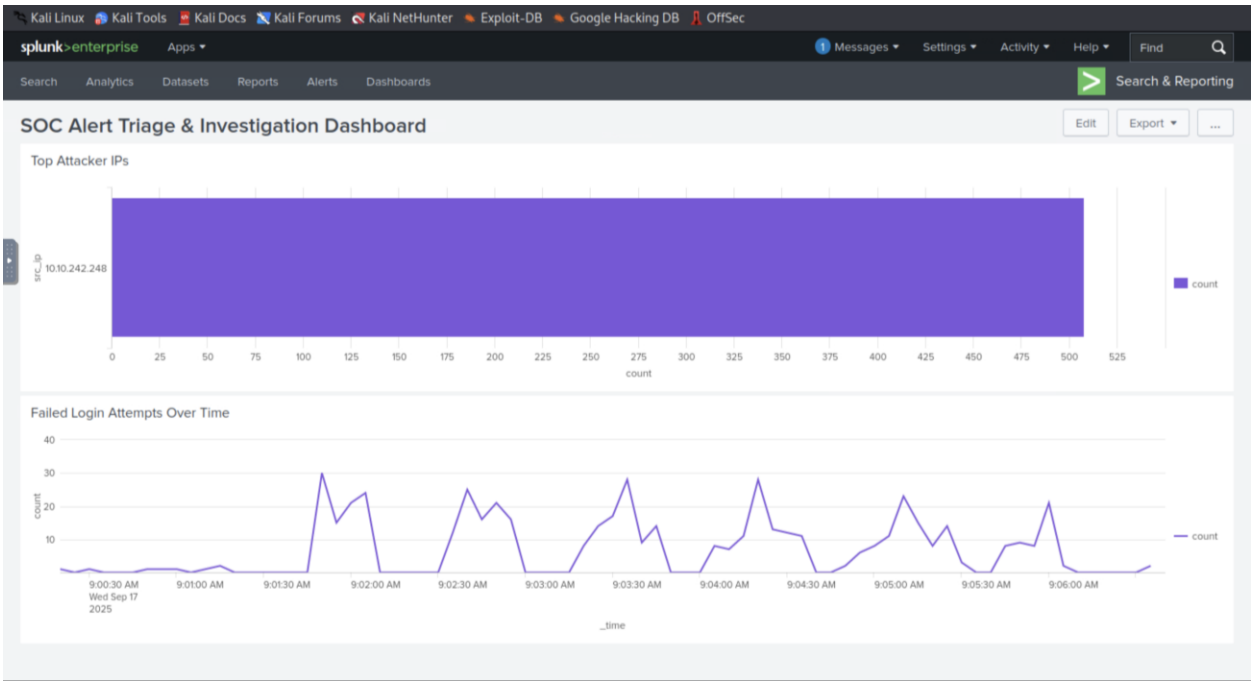

Figure 3. Time-series visualization of failed SSH login attempts showing repeated spikes in authentication failures during the investigation timeframe.

Key Observation:
The graph displays multiple rapid spikes in failed login attempts within short time intervals, which is consistent with automated brute-force attack activity rather than normal user behavior. Time-based analysis is important in SOC investigations because it allows analysts to correlate attack behavior with system events, authentication logs, and potential compromise timelines.

4. Identifying Targeted User Accounts

The investigation next focused on identifying which usernames were most frequently targeted during the suspicious SSH authentication activity. Analyzing targeted accounts helps determine attacker intent and identify accounts at higher risk of compromise.

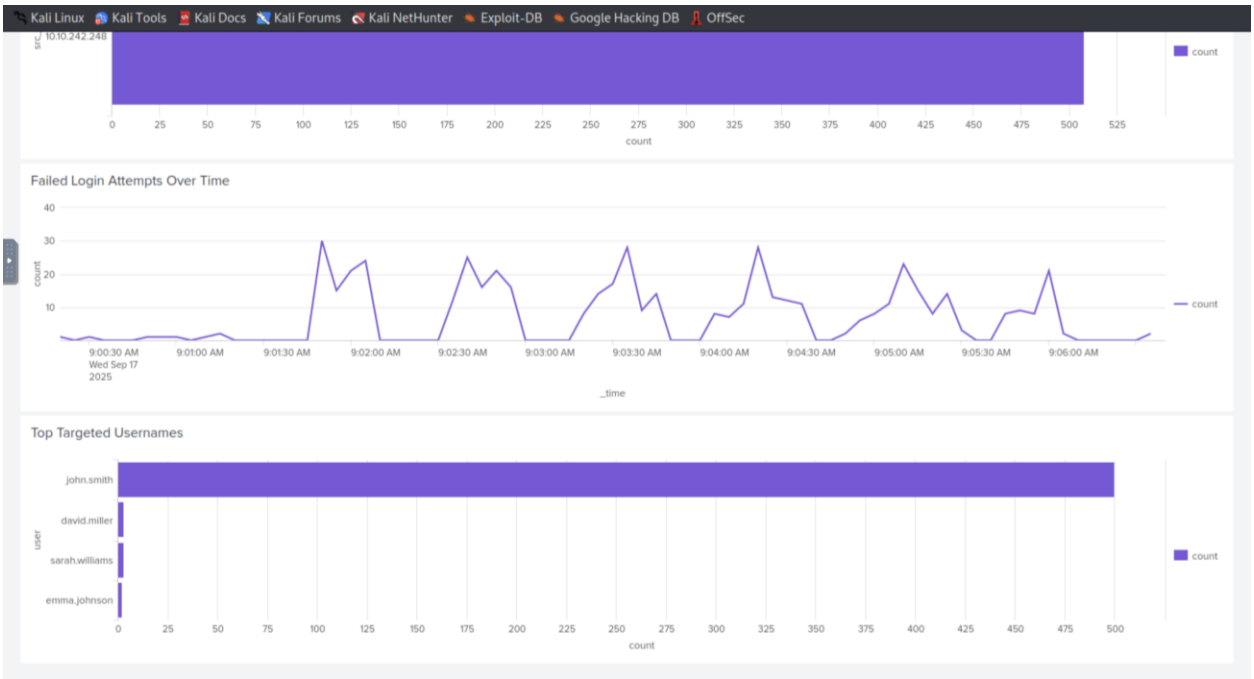

Figure 4. Dashboard visualization showing the usernames most frequently targeted during failed SSH login attempts.

Key Observation:
The account john.smith was overwhelmingly targeted compared to other usernames, suggesting the attacker may have been attempting credential stuffing or brute-force attacks against a high-value user account. Monitoring targeted usernames allows SOC analysts to prioritize account protection measures such as password resets, MFA enforcement, account lockout policies, and user activity reviews.

5. Detecting Successful SSH Logins

After analyzing failed authentication attempts, the investigation focused on identifying whether any login attempts were ultimately successful. Detecting successful logins following repeated failed attempts can indicate a possible account of compromise or attacker access into the system.

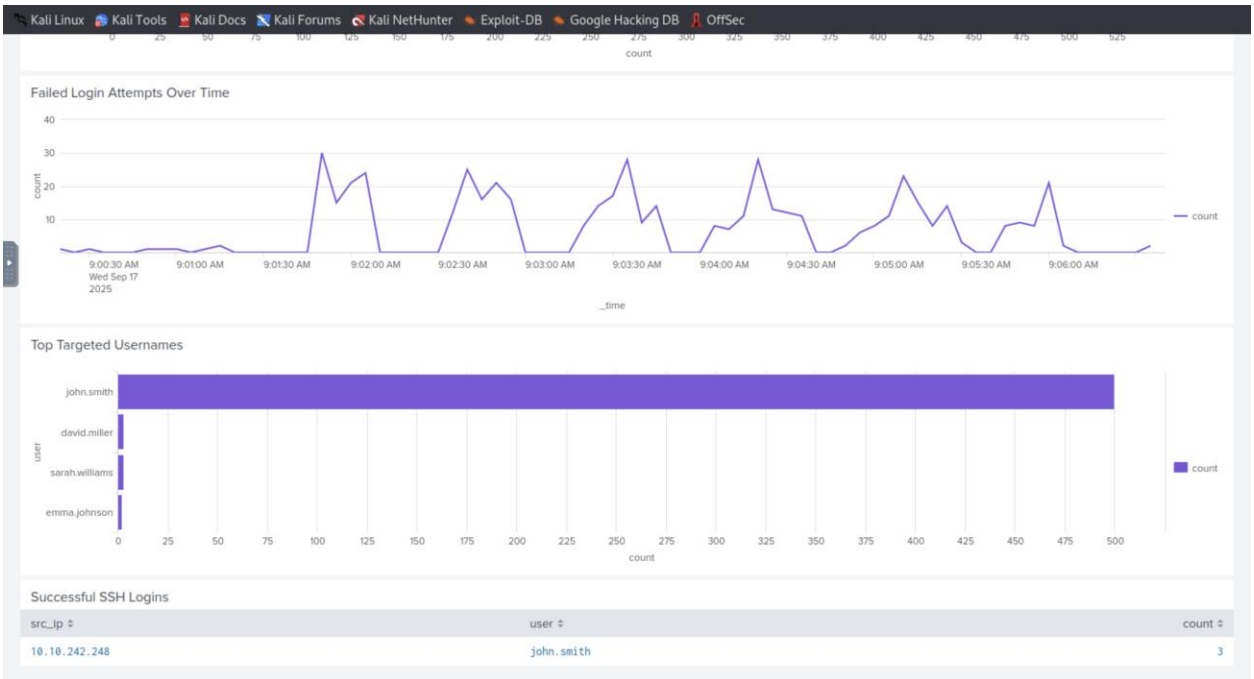

Figure 5. Dashboard table displaying successful SSH login events associated with suspicious authentication activity.

Key Observation:
The source IP address 10.10.242.248 successfully authenticated to the account john.smith three times after generating hundreds of failed login attempts, strongly suggesting a potential brute-force compromise. Correlating failed and successful login events is an important SOC investigation technique because it helps identify accounts that may have been compromised after repeated attack attempts.

6. Investigating Suspicious Web Requests

The investigation expanded beyond SSH authentication activity to examine suspicious web requests captured in HTTP logs. Analyzing unusual URI paths and response codes helps identify reconnaissance attempts, malicious scanning behavior, and potential exploitation activity targeting web applications.

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Suspicious Web Requests

_time	src_ip	uri_path	status	host
2025-09-14 22:07:19		/password.php	404	ce-splunk
2025-09-14 22:07:19		/form.html	404	ce-splunk
2025-09-14 22:07:18		/wp-admin/admin-ajax.php	200	ce-splunk
2025-09-14 22:07:18		/1.php	404	ce-splunk
2025-09-14 22:07:18		/upl.php	404	ce-splunk
2025-09-14 22:07:18		/systembc/password.php	404	ce-splunk
2025-09-14 22:05:17		/wp-admin/admin-ajax.php	200	ce-splunk
2025-09-14 22:04:22		/wp-corn.php	404	ce-splunk
2025-09-14 22:04:17		/wp-admin/admin-ajax.php	200	ce-splunk
2025-09-14 22:04:12		/wp-corn.php	404	ce-splunk
2025-09-14 22:04:06		/wp-corn.php	404	ce-splunk
2025-09-14 22:04:01		/wp-corn.php	404	ce-splunk
2025-09-14 21:31:51		/wp-admin/admin-ajax.php	200	ce-splunk
2025-09-14 21:26:28		/wp-corn.php	404	ce-splunk
2025-09-14 21:23:19		/wp-admin/theme-editor.php	200	ce-splunk
2025-09-14 21:22:46		/wp-admin/theme-editor.php	200	ce-splunk
2025-09-14 21:22:01		/wp-admin/admin-ajax.php	200	ce-splunk
2025-09-14 21:21:52		/wp-admin/load-styles.php	200	ce-splunk
2025-09-14 21:21:50		/wp-admin/theme-editor.php	200	ce-splunk
2025-09-14 21:21:28		/wp-admin/load-styles.php	200	ce-splunk

« Prev 1 2 3 4 5 6 7 8 9 10 Next »

Figure 6. Splunk dashboard table displaying suspicious web requests, targeted URI paths, and HTTP response codes associated with possible malicious activity.

Key Observation:
Multiple suspicious requests targeted sensitive web application paths such as /wp-admin/admin-ajax.php, /password.php, and /upl.php, which are commonly associated with web exploitation attempts, unauthorized access, or malicious scanning activity. The presence of repeated requests to WordPress administrative endpoints and abnormal PHP files may indicate automated reconnaissance tools or attempts to identify vulnerable web services within the environment.

7. Threat Severity Classification

To improve threat prioritization and SOC visibility, detected events were categorized into severity levels based on suspicious behavior patterns and associated indicators. Threat severity classification helps analysts quickly identify which activities require immediate investigation and response.

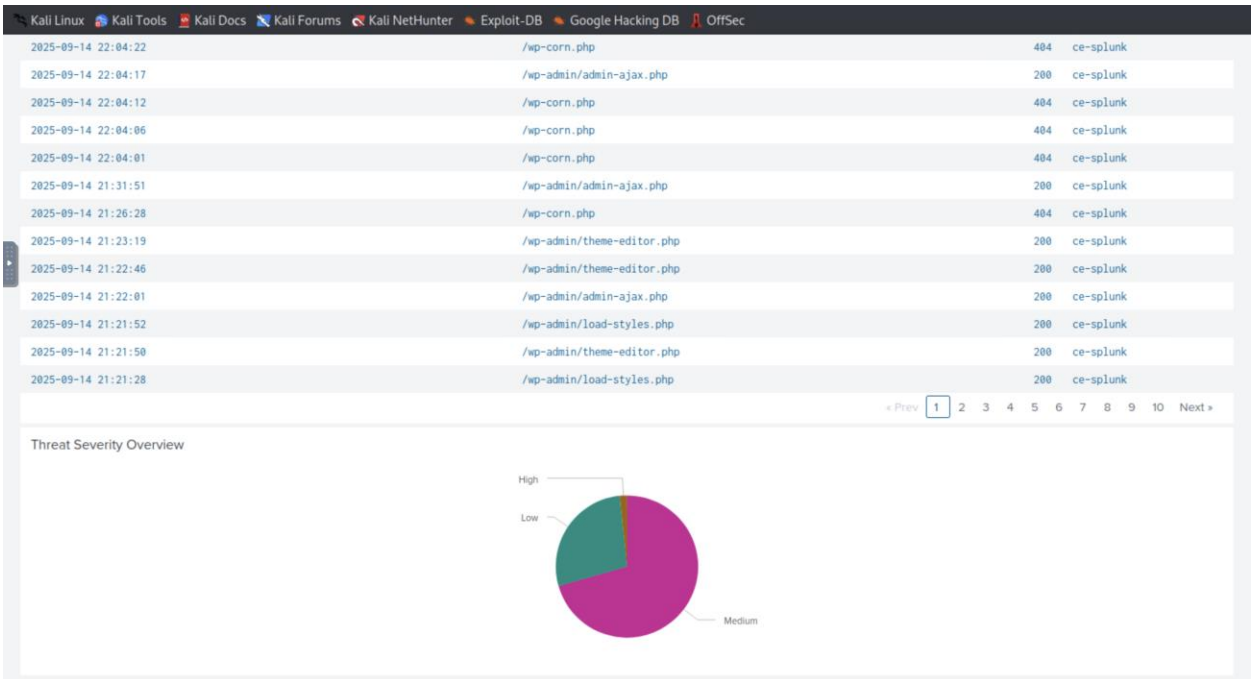

Figure 7. Threat severity overview displaying categorized event distribution across low, medium, and high severity levels within the monitored environment.

Key Observation:
Most detected events were categorized as medium severity, indicating a large amount of suspicious activity requiring investigation, while a smaller percentage of events were classified as high severity due to indicators associated with possible compromise attempts. Threat severity visualization allows SOC analysts to prioritize incident response efforts more effectively and focus resources on activities with the highest potential impact to the environment.

8. Correlating Failed and Successful Authentication Activity

To determine whether suspicious authentication attempts resulted in potential system compromise, failed and successful login events were correlated within the dashboard. This analysis helps identify attack patterns where repeated brute-force attempts eventually lead to unauthorized access.

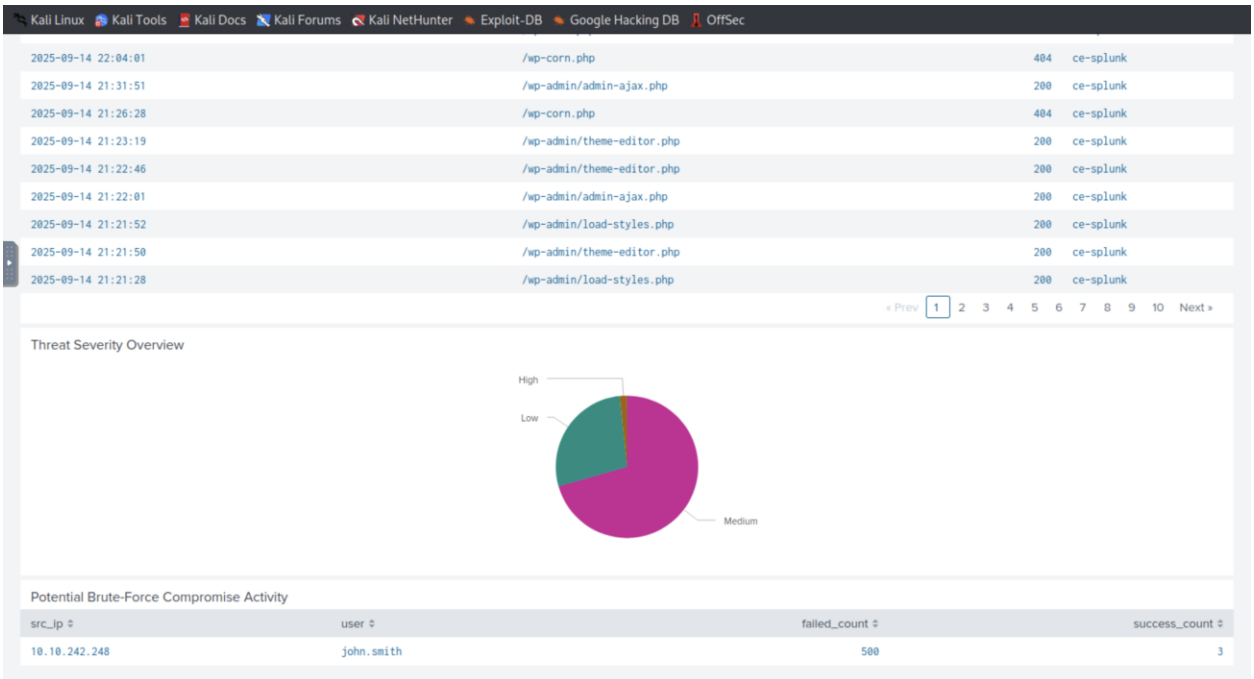

Figure 8. Correlated authentication activity showing the relationship between failed login attempts and successful SSH logins associated with the same user account and source IP address.

Key Observation:
The account john.smith experienced approximately 500 failed login attempts followed by 3 successful logins from the same source IP address, indicating a high probability of brute-force compromise activity. Correlation analysis is a critical SOC investigation technique because it enables analysts to connect isolated events into a larger attack narrative and identify indicators of account compromise more effectively.

9. Web Enumeration and Reconnaissance Activity Analysis

This section focuses on identifying suspicious web enumeration activity and potential attack behavior targeting web application resources. By analyzing requested web paths and authentication-related requests, the dashboard highlights commonly targeted administrative and login endpoints frequently associated with automated scanning and brute-force attacks.

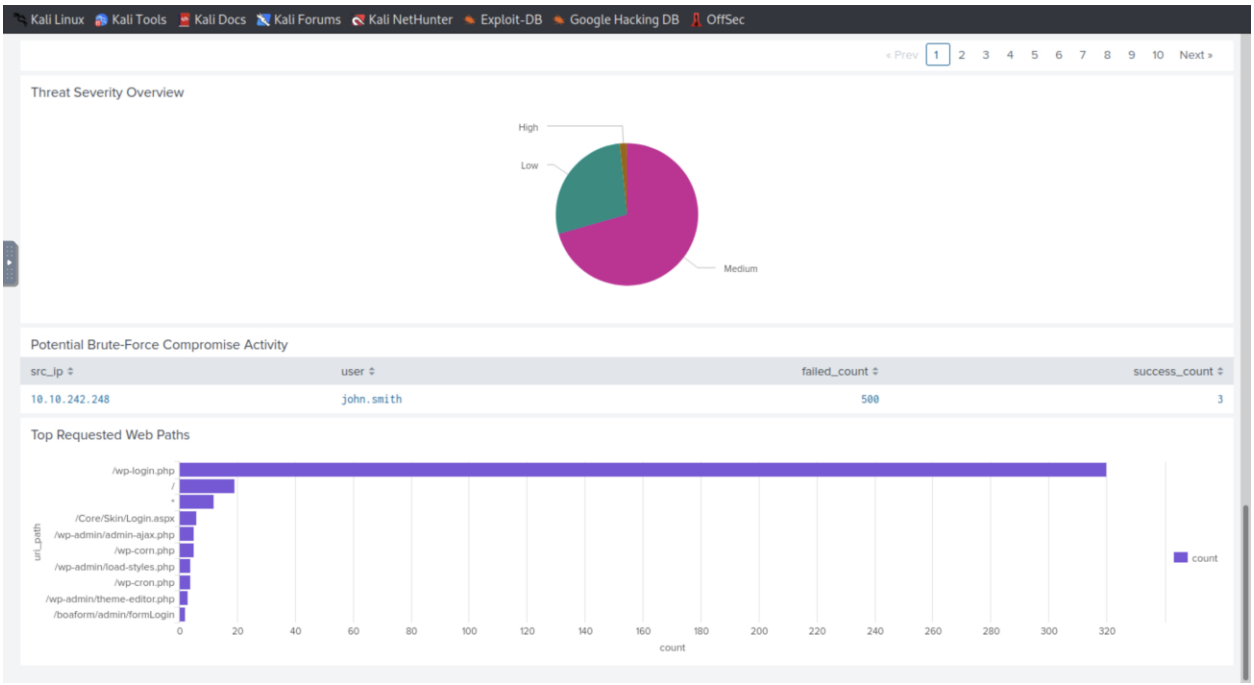

Figure 9. Dashboard panels showing threat severity distribution, brute-force compromise correlation, and the most frequently requested web application paths during suspicious activity.

Key Observation:

The /wp-login.php endpoint was the most frequently requested web path by a significant margin, suggesting automated attempts to access or brute-force WordPress authentication pages. Additional requests targeting administrative resources such as /wp-admin/admin-ajax.php and /wp-admin/theme-editor.php further indicate reconnaissance or exploitation attempts against the web application.

The dashboard also correlates failed authentication activity with successful login events, helping identify possible account compromise scenarios. The source IP 10.10.242.248 generated approximately 500 failed login attempts before recording successful authentication activity associated with the john.smith account.

The threat of severity visualization shows that most detected events were classified as medium severity, while a smaller number were categorized as low or high severity. This distribution suggests sustained suspicious activity that warrants continued monitoring and investigation by SOC analysts.

Visualizing attack paths and authentication trends within a centralized SIEM dashboard enables faster threat detection, improves incident triage efficiency, and supports proactive security monitoring practices.

10. HTTP Response and Web Request Behavior Analysis

This section analyzes web request behavior and HTTP response code distribution to identify indicators of reconnaissance, automated scanning, and possible exploitation attempts against web application resources. Monitoring requested paths alongside HTTP status responses helps security analysts determine attacker behavior and identify vulnerable or targeted services.

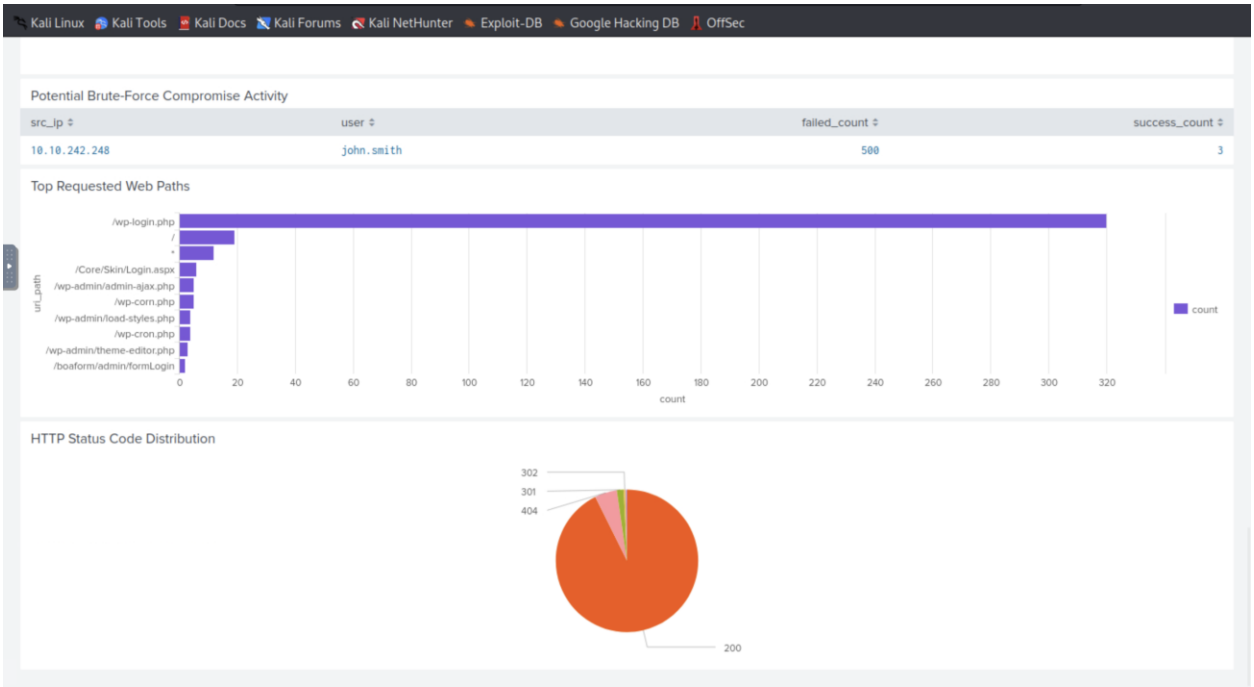

Figure 10. Dashboard panels displaying the most frequently requested web paths and the distribution of HTTP response status codes generated during suspicious web activity.

Key Observation:
The /wp-login.php endpoint generated the highest number of requests, significantly exceeding all other web paths. This behavior strongly suggests automated login attempts or brute-force activity targeting WordPress authentication systems.

Additionally, a large portion of the observed web requests returned HTTP status code 200, indicating that the requested resources were successfully accessible on the target web server.

Several additional administrative and application-related endpoints were targeted during the investigation, including:

- /wp-admin/admin-ajax.php
- /wp-admin/theme-editor.php

- /wp-cron.php
- /Core/Skin/Login.aspx

The presence of these requests suggests reconnaissance activity focused on discovering accessible administrative interfaces and potentially vulnerable web application components.

The HTTP status code chart further reveals a mixture of:

- 200 successful responses,
- 404 resource not found responses,
- 301 and 302 redirect responses.

This combination commonly appears during automated enumeration or vulnerability scanning campaigns, where attackers probe multiple paths to identify valid resources and login portals.

Analyzing web request patterns and HTTP response behavior within Splunk dashboards improves visibility into attacker reconnaissance techniques and supports faster identification of suspicious web activity during SOC investigations.

11. Persistence and System Modification Activity Investigation

This section focuses on identifying persistence-related behavior and suspicious system modifications observed during the SOC investigation. Persistence mechanisms are commonly used by attackers to maintain long-term access to compromised systems after initial intrusion attempts. The dashboard highlights abnormal account creation events, scheduled task activity, and suspicious file modification behavior detected through Sysmon logging.

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Potential Persistence Activity

_time	host	src_ip	uri_path	_raw
2025-09-17 09:12:10.914	ce-splunk			2025-09-17T09:12:10.914743+00:00 tryhackme-2404 useradd[3430]: new user: name=system-utm, UID=1002, GID=1002, home=/home/system-utm, shell=/bin/bash, from=/dev/pts/3
2025-09-17 09:05:01.106	ce-splunk			2025-09-17T09:05:01.106426+00:00 tryhackme-2404 CRON[2968]: pam_unix(cron:session): session closed for user root
2025-09-17 09:05:01.100	tryhackme-2404			2025-09-17T09:05:01.100939+00:00 tryhackme-2404 CRON[2969]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2025-09-17 09:05:01.098	ce-splunk			2025-09-17T09:05:01.098866+00:00 tryhackme-2404 CRON[2968]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2025-09-14 22:07:19.000	ce-splunk		/form.html	68.183.47.68 - - [14/Sep/2025:22:07:19 +0000] "GET /form.html HTTP/1.1" 404 42034 "-" "curl/8.1.2"
2025-09-14 21:19:22.000	ce-splunk		/wp-cron.php	171.251.232.50 - - [14/Sep/2025:21:19:22 +0000] "POST /wp-cron.php?doing_wp_cron=1751059162.9661519527435302734375 HTTP/1.1" 200 259 "-" "WordPress/6.8.1; http://171.251.232.50"
2025-09-14 20:58:16.000	ce-splunk		/wp-cron.php	171.251.232.50 - - [14/Sep/2025:20:58:16 +0000] "POST /wp-cron.php?doing_wp_cron=1751057896.4365489482879638671875 HTTP/1.1" 200 259 "-" "WordPress/6.8.1; http://171.251.232.50"
2025-09-14 19:25:25.000	ce-splunk		/wp-cron.php	171.251.232.50 - - [14/Sep/2025:19:25:25 +0000] "POST /wp-cron.php?doing_wp_cron=1751052325.5132489204406738281250 HTTP/1.1" 200 259 "-" "WordPress/6.8.1; http://171.251.232.50"
2025-09-14 16:55:41.000	ce-splunk		/wp-cron.php	171.251.232.50 - - [14/Sep/2025:16:55:41 +0000] "POST /wp-cron.php?doing_wp_cron=1751043341.2192239761352539062500 HTTP/1.1" 200 259 "-" "WordPress/6.8.1; http://171.251.232.50"
2025-08-30 09:44:48.000	WIN-H015			08/30/2025 09:44:48 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=2 EventType=4 ComputerName=WIN-H015 User=NOT_TRANSLATED Sid=S-1-5-18 SidType=0 SourceName=Microsoft-Windows-Sysmon Type=Information

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2025-08-30 09:44:48.000	WIN-H015	08/30/2025 09:44:48 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=2 EventType=4 ComputerName=WIN-H015 User=NOT_TRANSLATED Sid=S-1-5-18 SidType=0 SourceName=Microsoft-Windows-Sysmon Type=Information RecordNumber=5457 Keywords=None TaskCategory=File creation time changed (rule: FileCreateTime) OpCode=Info Message=File creation time changed: RuleName: - UtcTime: 2025-08-30 09:44:48.288 ProcessGuid: {c5d2b969-c7f6-68b2-6f01-000000002101} ProcessId: 7332 Image: C:\Windows\system32\msiexec.exe TargetFilename: C:\Program Files\SplunkUniversalForwarder\etc\system\bin\journald.sh CreationUtcTime: 2025-02-20 17:58:22.000 PreviousCreationUtcTime: 2025-08-30 09:44:48.288 User: NT AUTHORITY\SYSTEM
2025-08-30 09:44:47.000	WIN-H015	08/30/2025 09:44:47 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=2 EventType=4 ComputerName=WIN-H015 User=NOT_TRANSLATED Sid=S-1-5-18 SidType=0 SourceName=Microsoft-Windows-Sysmon Type=Information RecordNumber=5271 Keywords=None TaskCategory=File creation time changed (rule: FileCreateTime)

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2025-08-30 09:44:47.000	WIN-H015	08/30/2025 09:44:47 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=2 EventType=4 ComputerName=WIN-H015 User=NOT_TRANSLATED Sid=S-1-5-18 SidType=0 SourceName=Microsoft-Windows-Sysmon Type=Information RecordNumber=5271 Keywords=None TaskCategory=File creation time changed (rule: FileCreateTime) OpCode=Info Message=File creation time changed: RuleName: - UtcTime: 2025-08-30 09:44:47.006 ProcessGuid: {c5d2b969-c7f6-68b2-6f01-000000002101} ProcessId: 7332 Image: C:\Windows\system32\msiexec.exe TargetFilename: C:\Program Files\SplunkUniversalForwarder\share\splunk\3rdparty\Copyright-for-curl-8.5.0.txt CreationUtcTime: 2025-02-20 18:04:10.000 PreviousCreationUtcTime: 2025-08-30 09:44:47.006 User: NT AUTHORITY\SYSTEM
2025-08-30 09:44:46.000	WIN-H015	08/30/2025 09:44:46 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=2 EventType=4 ComputerName=WIN-H015 User=NOT_TRANSLATED Sid=S-1-5-18 SidType=0 SourceName=Microsoft-Windows-Sysmon Type=Information RecordNumber=5208 Keywords=None

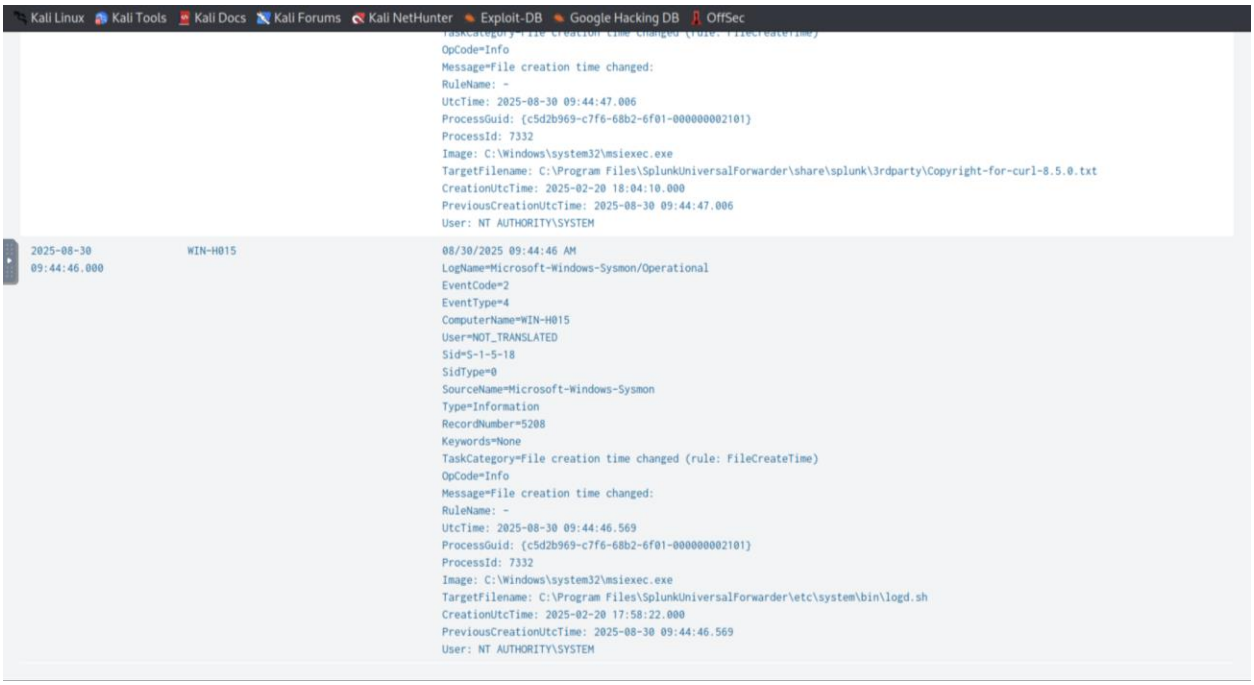


Figure 11. Dashboard logs showing suspicious persistence-related events, including unauthorized account creation, cron activity, and system file modification events captured through Sysmon monitoring.

Key Observation:
Several events indicate potential persistence activity on monitored systems. Notable findings include the creation of a suspicious user account named system-utm, repeated cron-related execution activity, and multiple file creation time modification events associated with msiexec.exe on the Windows host WIN-H015.

The Linux system logs revealed the following suspicious event:

- useradd[3430]: new user: name=system-utm

The creation of unexpected accounts may indicate an attacker attempting to establish persistence or maintain unauthorized access to the environment.

Additional cron-related activity was also identified, including scheduled command execution events involving root privileges. Attackers commonly abuse cron jobs to automate malicious payload execution or re-establish persistence after system reboots.

The Windows Sysmon logs further showed multiple EventCode=2 entries associated with:

- TaskCategory=File creation time changed
- Image: C:\Windows\system32\msiexec.exe

The events referenced several modified files within Splunk Universal Forwarder directories, including:

- journald.sh
- logd.sh
- Copyright-for-curl-8.5.0.txt

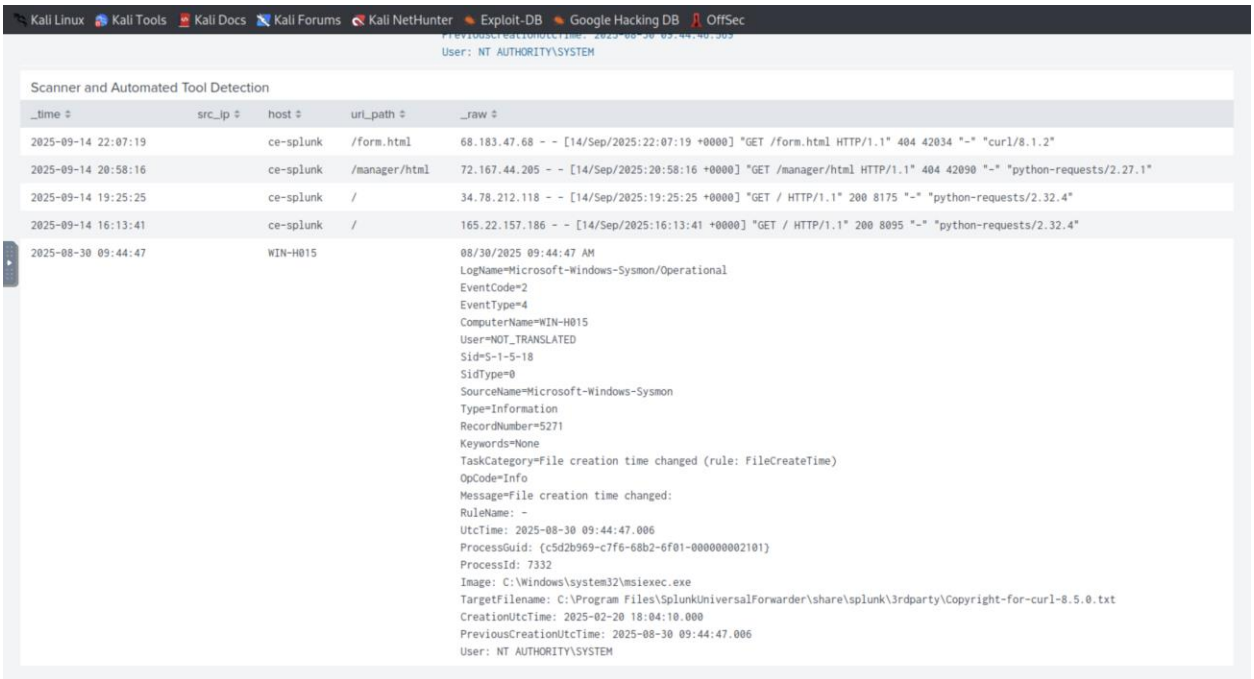
Timestamp modification behavior may indicate attempts to alter forensic evidence or disguise malicious activity by changing file metadata.

Persistence-related indicators are critical during incident response investigations because they may reveal how attackers maintain continued access within an environment. Monitoring account creation activity, scheduled tasks, and abnormal file modifications help SOC analysts detect stealth techniques commonly associated with post-compromise behavior.

Correlating Linux authentication logs, cron activity, and Sysmon telemetry within Splunk provides valuable visibility into persistence techniques and strengthens proactive threat detection capabilities during SOC investigations.

12. Automated Scanner and Tool Detection Analysis

This section focuses on identifying activity generated by automated scanning tools and reconnaissance frameworks. Automated scanners are commonly used by attackers to discover accessible resources, identify vulnerable services, and enumerate web applications prior to exploitation attempts. The dashboard highlights suspicious request patterns, user-agent behavior, and scanning-related HTTP activity observed during the investigation.



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User: NT AUTHORITY\SYSTEM

Scanner and Automated Tool Detection

_time	src_ip	host	uri_path	_raw
2025-09-14 22:07:19		ce-splunk	/form.html	68.183.47.68 - - [14/Sep/2025:22:07:19 +0000] "GET /form.html HTTP/1.1" 404 42034 "-" "curl/8.1.2"
2025-09-14 20:58:16		ce-splunk	/manager/html	72.167.44.205 - - [14/Sep/2025:20:58:16 +0000] "GET /manager/html HTTP/1.1" 404 42090 "-" "python-requests/2.27.1"
2025-09-14 19:25:25		ce-splunk	/	34.78.212.118 - - [14/Sep/2025:19:25:25 +0000] "GET / HTTP/1.1" 200 8175 "-" "python-requests/2.32.4"
2025-09-14 16:13:41		ce-splunk	/	165.22.157.186 - - [14/Sep/2025:16:13:41 +0000] "GET / HTTP/1.1" 200 8095 "-" "python-requests/2.32.4"
2025-08-30 09:44:47		WIN-H015		08/30/2025 09:44:47 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=2 EventType=4 ComputerName=WIN-H015 User=NOT_TRANSLATED Sid=S-1-5-18 SidType=0 SourceName=Microsoft-Windows-Sysmon Type=Information RecordNumber=5271 Keywords=None TaskCategory=File creation time changed (rule: FileCreateTime) OpCode=Info Message=File creation time changed: RuleName: - UtcTime: 2025-08-30 09:44:47.006 ProcessGuid: {c5d2b969-c7f6-68b2-6f01-000000002101} ProcessId: 7332 Image: C:\Windows\system32\msiexec.exe TargetFilename: C:\Program Files\SplunkUniversalForwarder\share\splunk\3rdparty\Copyright-for-curl-8.5.0.txt CreationUtcTime: 2025-02-20 18:04:10.000 PreviousCreationUtcTime: 2025-08-30 09:44:47.006 User: NT AUTHORITY\SYSTEM

Figure 12. Dashboard results showing suspicious web requests and indicators of automated reconnaissance activity generated by scanning tools and scripted HTTP requests.

Key Observation:
Multiple web requests were associated with automated tools such as curl and python-requests, indicating likely reconnaissance or scripted scanning behavior against the monitored web server.

The dashboard identified several suspicious requests targeting sensitive or administrative web paths, including:

- /form.html
- /manager/html
- /

The requests originated from multiple external IP addresses, including:

- 68.183.47.68
- 72.167.44.205
- 34.78.212.118
- 165.22.157.186

The HTTP request logs also revealed the use of automated user agents such as:

- curl/8.1.2
- python-requests/2.27.1

- python-requests/2.32.4

The use of command-line tools and Python request libraries is commonly associated with automated enumeration, vulnerability scanning, or scripted attack activity rather than normal user browsing behavior.

One request targeting /manager/html returned a 404 response code, suggesting the scanner was probing for administrative interfaces that may exist on vulnerable systems.

Automated reconnaissance activity is often an early indicator of attempted intrusion or vulnerability assessment by attackers. Detecting unusual request patterns and scripted user agents helps SOC analysts identify suspicious activity before successful exploitation occurs.

Monitoring automated scanning behavior through Splunk dashboards strengthens early threat detection capabilities and provides valuable visibility into attacker reconnaissance techniques targeting enterprise systems.

Conclusion

The Splunk dashboard successfully identified multiple indicators of suspicious and potentially malicious activity, including brute-force SSH attacks, suspicious web enumeration, persistence-related events, and automated scanning behavior. Correlation of authentication logs, web requests, Sysmon telemetry, and severity classifications improved visibility into attacker behavior and supported efficient SOC investigation workflows.

Overall, the dashboard demonstrates how SIEM platforms such as Splunk can assist SOC analysts in threat detection, event correlation, incident triage, and proactive security monitoring within enterprise environments.